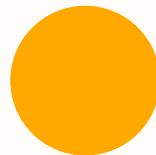




Networking Intro

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Networking

- We have a rough idea of how computers work
 - Generally
- Your computer does a lot
- But it doesn't do everything
- Many things we serve out to other computers





Networking

- We have other computers do work for us
 - More powerful
- But how can these computers communicate
- **Networking**
 - Connect devices together





Networking

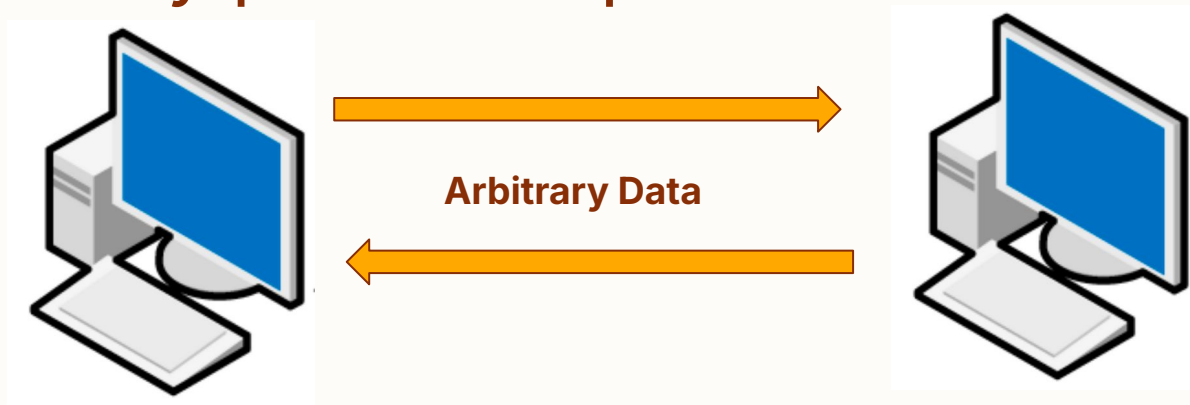
- Just like most things
 - Communication is often just a set of standards
- Networking Protocols
- Before low level protocols
 - Let's talk big picture
- Mental Models for communication





Networking

- Networking just connects two device
- Doesn't say anything about relation or any particular protocol



Networking

- We often think about these as computers
- Not limited to computers
- Any device





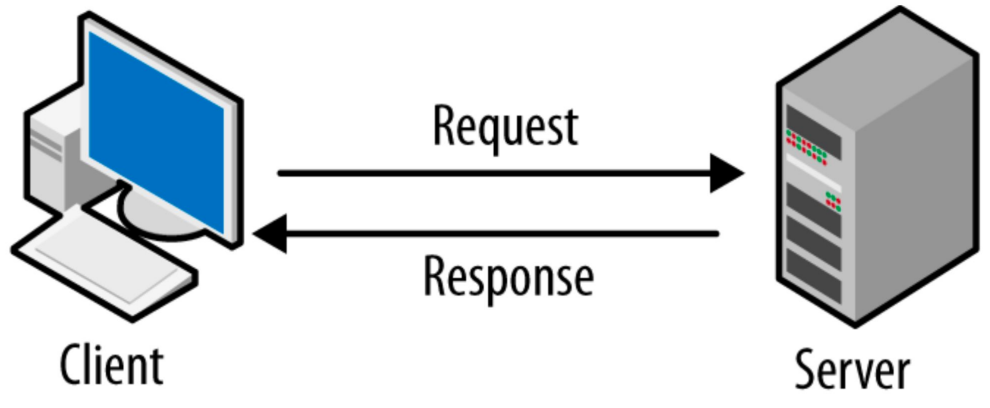
Networking

- Networking is often employed to offload work
 - To a more well suited device
- More powerful computer
- Something with different hardware
- Scanners, printers, etc



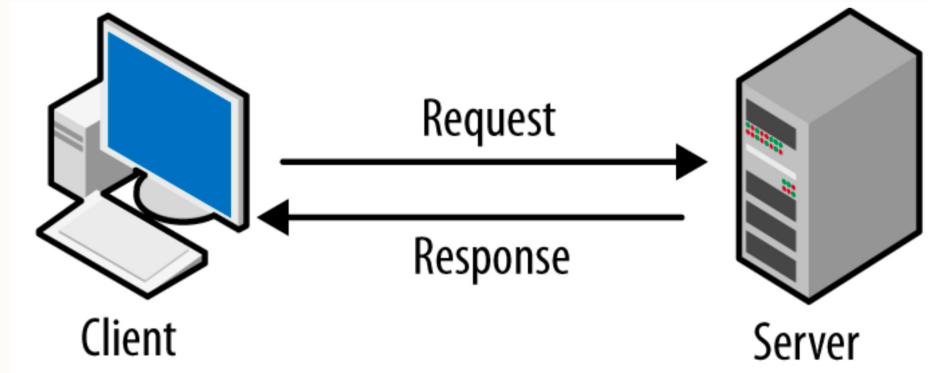
Networking

- Client-Server model
 - The most important mental model for networking



Networking

- Client:
 - sends a "request" to server
- Server:
 - Sends a "response" to the client





Networking

- Server is more powerful at doing a particular task
- Or it is simply purpose built
 - Only way of accessing some data
- Client is usually **"you"**
 - Your computer
 - Your device





Networking

- Visiting a web page is an example of this
- You (client) requests data.
- Website (server) responds with the HTML
- Some tasks in general need a beefier computer → LLMs





Networking

- Keep in mind
- Client-Server is not the only way two devices can communicate over a network
- Just the most common
- Any other examples of this relation?



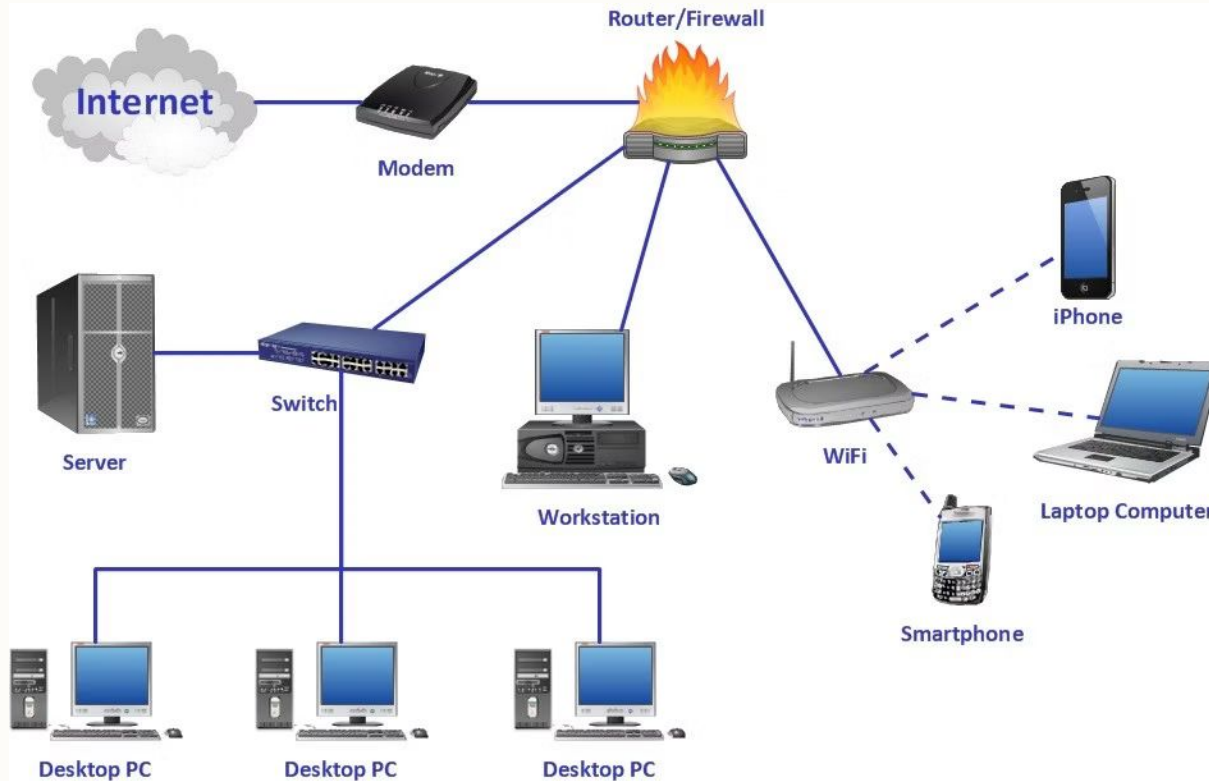


Networking

- A network is just a bunch of interconnected devices
- A network consists of **many devices** communicating
- Your home networking might look something like:



Networking





Networking

- Internet is all done thru a network
- Specifically something called HTTP
 - See URLs
- Usually client-server relationships
- HTTP is only **one** standard
 - Common on the internet*
- Let's break that down a bit





Networking

- HTTP
- **Hypertext Transfer Protocol**
- Standard used to communicate all kinds of data
 - Translated down into the bit level
- Primarily used by browsers





Networking

- You enter some URL
- Your browser send **HTTP requests**
 - To a server
- Server replies via HTTP (often with HTML) for your browser to render
- By default HTTP is unencrypted





Networking

- **HTTPS** is a more common standard nowadays
- **S** stands for secure
- Tells your browser to be sure to encrypt your data before sending it OTA (or where ever)





Networking

- HTTP(S) is broken down into what they call
- **HTTP Methods**
- Just a way to organize requests by their primary purpose
- Set up by the standard
- We will talk about the most important ones





Networking

- HTTP **GET**
- Client sends a **GET** request
- **Retrieve data from the server,**
HTML usually
- Read only (*)
- Most common HTTP method





Networking

- **HTTP POST**
- Client sends a **POST** request
 - **Sends additional data**
- Submit data and create a **new** resource on the server
- Think "account creation"





Networking

- **HTTP PUT**
- Client sends a **PUT request**
 - **Sends additional data**
- Replace existing resource completely on the server
- Think "replace an entire user"





Networking

- **HTTP PATCH**
- Client sends a **PATCH request**
 - **Sends additional data**
- Partially update an existing resource on the server
- Think "update the user's password"





Networking

- **HTTP DELETE**
- Client sends a **DELETE request**
 - **Sends additional data**
- Completely delete an existing resource on the server
- Think "delete the user entirely"





Networking

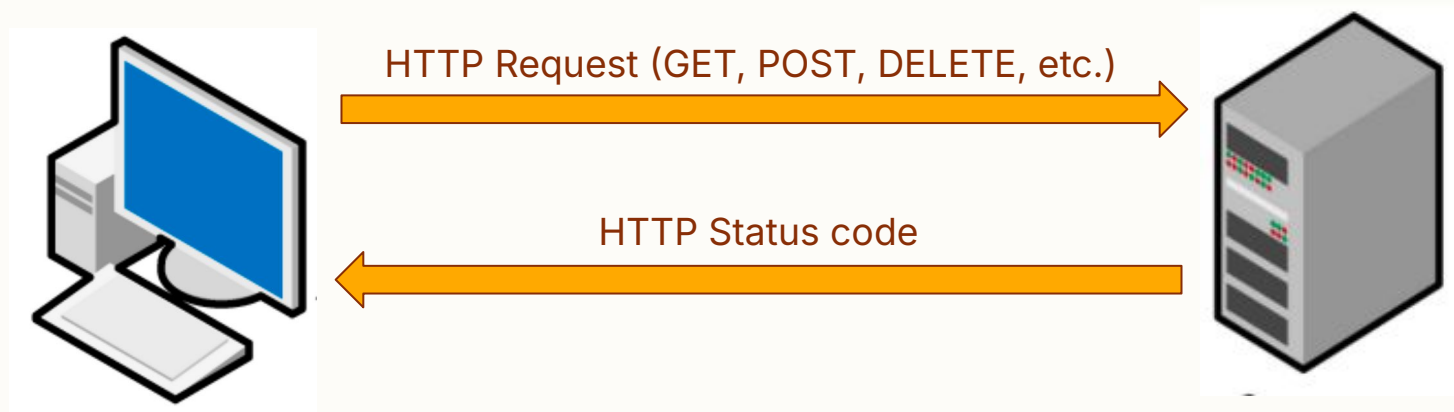
- HTTP methods are usually sent in a request by the client
- But that's only half of the equation
- HTTP is bi-directional communication standard
- What lives on the other end?





Networking

- Usually, the server replies with a special status code
- These codes are standardized





Networking

- Status codes exist to tell the client the state of their request
- Did it fail?
- Succeed?
- Is the server busy?
- Does that resource even exist?
- How are they organized?



Networking

- 100 - 599 *
- First digit tells you the category

HTTP Status Codes



Networking

- 100 Range
- Informational
- Rarely something you need to worry about
- More useful for browsers

HTTP Status Codes



- *100 Continue*
- *102 Processing*
- *103 Switching Protocols*
- *etc.*

Networking

- 200 Range
- Success Codes
- You are hoping for these
- **200 Success**
- **201 Created**
- **202 Accepted**

HTTP Status Codes



Networking

- 300 Range
- Redirection Codes
- Sends you to another place
- **301 Moved Permanently**
- **302 Found**
- etc.

HTTP Status Codes



Networking

- 400 Range
- Client Error Codes
- The client made a mistake
- **400 Bad Request**
- **401 Unauthorized**
- ***404 Not found***

HTTP Status Codes



Networking

- 500 Range
- Server Error Codes
- The server made a mistake
- **500 Internal server error**
- **505 HTTP version not supported**

HTTP Status Codes





Networking

- There are more status codes than we've covered
- But not as many as you'd think
- Not all numbers are reserved
- Let's peek at some docs
- HTTP Status 418
 - I'm a teapot





Networking

- REST API
- API = Application Program Interface
- REST = Representational State Transfer
- An http interface for accessing a program (usually a server)





Networking

- Insomnia Simulator
- Will be used for your lab next week
- HTTP testing simulator
- Receive JSON as a response usually
- Let's talk about that





Networking

- Each device is assigned an **IP** address
- **Unique device network ID**
- ~ 24.156.99.202
- You send request to an IP
- Get responses from an IP





Networking

- But what happens when I visit
- <https://www.google.com>
- That turned into some kind of IP like
- 24.156.99.202
- What happened?





Networking

- Domain Name Service
- Resolves Web Addresses into IP Address
- Distributed System
- Essential for communication in the modern sense
- Nslookup for going backwards



Networking

